

Saehyun Choi

Assistant Professor

Department of Chemistry and Nanoscience

Ewha Womans University

saehyun.choi@ewha.ac.kr (Last updated: 2026-8-21)

(82) 02-3277-2321

<https://sites.google.com/view/the-choi-group-ewha>

EDUCATION

Ph.D., Chemistry

September 2016 – August 2022

The Pennsylvania State University (PA, United States)

Advisor: Professor Christine D. Keating

M.S., Chemistry and Nanoscience

March 2013 – February 2015

Ewha Womans University (Seoul, South Korea)

Advisor: Professor Jun Soo Kim

B.A., Chemistry and Nanoscience, Minor in Life Science

March 2009 – February 2013

Ewha Womans University (Seoul, South Korea)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

October 2022 – February 2026

Department of Chemistry

University of California, Berkeley (CA, United States)

Advisor: Prof. Jay T. Groves

Researcher

March 2015 – February 2016

Department of Chemistry and Nanoscience

Ewha Womans University (Seoul, South Korea)

Advisor: Prof. Jun Soo Kim

RESEARCH: Phase Transition Thermodynamics, Liquid-liquid Phase Separation (LLPS), Coacervates, Peptide/RNA chemistry, Artificial cells, Biomolecular Condensates, Advanced Imaging, Immune Cell Signaling

I. Postdoctoral research (September 2022 – February 2026):

- Investigating protein phase transition mechanisms of signaling proteins in protein condensates in live T cells.
- Developed single-molecule microscopy analysis tools using total internal reflection fluorescence (TIRF) microscopy.

II. Ph.D. research (November 2016 – August 2022):

Liquid-liquid phase separation (LLPS) of peptides as a model for prebiotic and biotic compartmentalization.

- Developed multiphasic coacervates composed of polypeptides that alter local RNA chemistry.
- Established the thermodynamic analysis to validate the altered RNA chemistry in multiphasic coacervates.
- Investigated the effect of polypeptide length on the local electrostatic microenvironments of coacervates.
- Incorporated coacervates into laboratory FeS hydrothermal vents.

PUBLICATIONS

1. A. J. Rubin, T. T. Dao, A. V. Schueppert, **S. Choi**, J. T. Groves, A. Regev, A. K. Shalek, Disordered protein LAT encodes relative levels of signaling pathways in T cell activation, *Science* 392, eads6847(2026).
2. **S. Choi**, S. P. Liu, M. O. Meyer, P. C. Bevilacqua, C. D. Keating, Primitive Molecular Buffering by Low-Multivalency Coacervates, *Journal of Physical Chemistry B* 129, 33, 8368–8381 (2025).
3. **S. Choi**⁺, A. R. Knoerdel⁺, C. E. Sing, C. D. Keating, Effect of polypeptide complex coacervate microenvironment on protonation of a guest molecule, *Journal of Physical Chemistry B* 127, 26, 5978–5991 (2023).
4. M. O. Meyer, R. Yamagami, **S. Choi**, C. D. Keating, P. C. Bevilacqua, RNA folding studies inside peptide-rich droplets reveal roles of modified nucleosides at the origin of life, *Science Advances* 9, eadh5152 (2023).
5. M. O. Meyer, **S. Choi**, C. D. Keating, P. C. Bevilacqua, R. Yamagami, Structure-seq of tRNAs and other short RNAs in droplets and *in vivo*, *Methods in Enzymology* 691, 81-126, (2023).
6. **S. Choi**, M. O. Meyer, P. C. Bevilacqua, C. D. Keating, Phase-specific RNA accumulation and duplex thermodynamics in multiphase coacervate models for membraneless organelles, *Nature Chemistry* 14, 1110–1117 (2022).
7. F. P. Cakmak, **S. Choi**⁺, M. O. Meyer⁺, P. C. Bevilacqua, C. D. Keating, Prebiotically-relevant low polyion multivalency can improve functionality of membraneless compartments, *Nature Communications* 11, 5949 (2020).
8. I. Oh, **S. Choi**, Y. Jung, J. S. Kim, Entropic effect of macromolecular crowding enhances binding between nucleosome clutches in heterochromatin, but not in euchromatin, *Scientific Reports* 8, 5469 (2018).
9. I. Oh, **S. Choi**, Y. Jung, J. S. Kim, Phase separation of a Lennard-Jones fluid interacting with a long, condensed polymer chain: implications to the nuclear body formation near chromosomes, *Soft Matter* 11, 6450 (2015).
10. **S. Choi**⁺, E. Jang⁺, J. S. Kim, In-layer stacking competition during ice growth, *Journal of Chemical Physics* 140, 014701 (2014).
11. I. Oh, **S. Choi**, Y. Jung, J. S. Kim, Unusual size-dependence of effective interactions between collapsed polymers in crowded environments, *Soft Matter* 10, 9098 (2014).
12. M. Seo⁺, E. Jang⁺, K. Kim, **S. Choi**, J. S. Kim, Understanding anisotropic growth behavior of hexagonal ice on a molecular scale: a molecular dynamics simulation study, *Journal of Chemical Physics* 137, 154503 (2012).

IN PREPARATION

1. **S. Choi**⁺, S. Kim⁺, T. J. Eisen, S. Morita, Jay T. Groves, Stimulation of CD28 or PD-1 Differentially Modulate the Propensity of LAT Condensate Nucleation by Activated T-cell Receptors, *invited submission in Philos. Trans. R. Soc. B* (2026).
2. **S. Choi**⁺, T. J. Eisen⁺, S. Morita, John Kuriyan, Jay T. Groves, Assembly of a Partial Signaling Unit Containing ITK and SLP76 Precedes T-cell Receptor Activation, *in preparation*.
3. **S. Choi**, T. J. Eisen, S. Morita, Jay T. Groves, Single-molecule Displacement Imaging Reveals Low-yield Maturation of TCR-induced LAT Prenucleation Clusters into Signaling Condensates, *in preparation*.
4. E. Mullikin⁺, **S. Choi**⁺, L. Barge, C. D. Keating, Interactions of Complex Coacervates and Precipitates Under Simulated Hydrothermal Vent Conditions, *in preparation*.

⁺ equally contributed

FUNDING & SCHOLARSHIPS

- Co-Investigator of Brain Pool + Program funded by National Research Foundation of Korea (2026-)
- Co-Investigator Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2026-).
- Ad Hoc Grant by Burroughs Wellcome Funds for the 2024 Gordon Research Seminar ‘Biointerface Science’ as a conference chair (2024)
- Group leader for a postdoctoral research team at UCB and LBNL, “Berkeley Innovative Green-Scientists”, funded by Korean Federation of Science and Technology Societies (2023)
- NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Graduate Fellowship (August 2019- August 2022)
- NASA Astrobiology Early Career Travel Award (Fall 2017)
- Research grant for undergraduate researchers by Korea Foundation for the Advancement of Science & Creativity, South Korea (2010)

HONORS & AWARDS

- Berkeley Postdoctoral Association Professional Development Awards (2024)
- Continuing Graduate Awards at Penn State (2021-2022)
- Lotte Scholarship, South Korea (2011-2012)
- High Honors Student of Ewha Womans Univ. (Spring 2009, Fall 2011 and Spring 2012)

PROFESSIONAL EXPERIENCES

- NASA Solar System Science (SCUBED) funding review panel (October 2025)
- NASA Exobiology Program (Prebiotic Chemistry) funding review panel (May 2024)
- Gordon Research Seminar elected chair for ‘Biointerface Science’ in 2022 and conference in June 2024 (June 2022- June 2024)
- Protocell Models by the Max Planck Institute of Molecular Cell Biology and Genetics, ‘the ProtoMet Innovative Training Network’, attendee (July 5-8, 2021)
- 17th Astrobiology Summer School by NASA and ESA, attendee (July 2019)
- NASA Jet Propulsion Lab, Collaboration travel to Dr. Laurie Barge (November 2018)
- Brain Korea 21, Research assistant (Spring 2013 – Spring 2016)

PRESENTATIONS

Oral presentations

- Gordon Research Seminar ‘Biointerface Sciences’, invited talk (June 2026)
- Summer symposium by Colloid and Molecular Assembly division, the Polymer Society of Korea, invited talk (June 2026)
- Chung-Ang University, Global SRC for Systems Chemistry, invited talk (June 2026)
- Winter Symposium by Division of Physical Chemistry, Korean Chemical Society, invited talk (February 2026)
- Penn State Univ, Department of Chemistry, PA, invited talk (July 2025)
- DGIST, South Korea at the Chemistry Department Seminar, invited talk (December 2024)
- Ewha Womans University, Department of Human-Machine-Bioengineering, invited talk (August 2023)

- International Conference on Biological Physics (August 2023)
- Gordon Research Seminar ‘Colloidal, Macromolecular and Polyelectrolyte Solutions’ (November 2022)
- PCE3 Seminar Series, invited talk (December 2021)
- ACS Spring annual meeting (virtual, April 2021)
- ACS annual meeting (virtual, Summer 2020)
- ChemBio Seminar at Penn State (Spring 2020)
- Korean Chemistry Society in Physical Chemistry (September 2013)

Poster Presentations

- Gordon Research Seminar and Conference ‘Biointerface Science’ (June 2026)
- Biophysical Society (BPS, February 2025)
- Gordon Research Seminar and Conference ‘Biointerface Science’ (June 2024)
- International Conference on Biological Physics (August 2023)
- Gordon Research Seminar and Conference ‘Biointerface Science’ (June 2022)
- Gordon Research Conference ‘Colloidal, Macromolecular and Polyelectrolyte Solutions’ (November 2022)
- 2021 MRS Fall meeting, finalist for the Poster Award (November 2021)
- 2021 AbGradCon, poster presentation and discussion leader (September 2021)
- Molecular Origins of Life, Munich 2021 (August 2021)
- Molecular Origins of Life, Munich 2020 (August 2020)
- 2019 AbSciCon conference, finalist of poster competition (June 2019)
- 92nd ACS colloid and surface symposium (April 2018)
- 2017 AbGradCon (June 2017)
- American Chemical Society (ACS, September 2014)

TEACHING AND MENTORING EXPERIENCES

- General Chemistry Lecture I, Ewha Womans University (Spring 2026)
- Physical Chemistry Lab, Ewha Womans University (Spring 2026)
- Mentoring two undergraduate students at UCB (Jessica R. Tijam, Chemistry Major and Erin Ortega, Molecular and Cell Biology Major) (Summer 2023 – Present)
- Guest lectures for Biophysical Chemistry (MCBC100A) at UC Berkeley (Fall 2023)
- Trained and maintained the confocal microscope in the Chemistry user facility and other spectroscopic instruments in the Keating group. Trained about 20 users per year (Summer 2018 – August 2022)
- Mentored the junior graduate students in the Keating group for laboratory skills (Fall 2020 - August 2022)
- Mentored the undergraduate student (Sindy Liu Peng, Chemistry Major at Penn State)
- Helped Sindy Peng for two conferences and three awarded undergraduate research fellowships (Summer 2020 – August 2022)
- Mentored the REU summer undergraduate student (Luis G. Ramíerz-Pagán, UPR) (Summer 2019)
- Teaching assistant (TA) for General Chemistry Lecture and Laboratory CHEM 110 & 111 (Penn State, Fall 2016)
- Tutoring for undergraduate students from General Chemistry Courses such as CHEM 110, 112, 111 and 113 (Penn State, Fall 2018)

- TA for General Chemistry Lecture and Laboratory CHEM 101 & 113 (Penn State, Spring 2017 – Spring 2018)
- Head TA for Physical Methods in Chemistry (Ewha Womans Univ., Spring 2014)
- TA for Physical Methods in Chemistry (Ewha Womans Univ., Spring 2013)
- Tutored for undergraduate students for Physical Chemistry Courses (Ewha Womans Univ., Fall 2011)

SCIENCE OUTREACH & OTHER ACTIVITIES

- Committee member for the Lawrence Berkeley National Laboratory MBIB annual meeting (March- April 2025)
- Mentor for Be-A-Scientist program, Community Resources for Science, Berkeley (Spring 2024 & Spring 2025)
- Invited panel in the panel discussion on “Time Management for MRSEC (an NSF-funded Materials Science Center) group of Graduate Research Experience and Transition to Grad School” at Penn State (July 28th 2021)
- Sub-committee member of One Penn State 2025 Leadership Team to develop the university-wide program to enhance equal opportunity and life-long education (Spring 2020 – Spring 2022)
- Member of the outreach team of MRSEC Penn State (Spring 2019– Summer 2020)
- Committee member for the professional development conference by Graduate Women in Science, State College Chapter at Penn State (May 2018– February 2020)
- Co-chair of Women in STEM Month in Graduate Women in Science, State College Chapter at Penn State (May 2018– May 2020)
- Co-booth leader for the science outreach demo titled “Painting with Lights” in Art-fest (State College, PA, Summer 2019)
- Co-invented the new science demo titled “Painting with Lights” in the outreach team of MRSEC Penn State with Dr. Michelle Tomczyk (Summer 2019)
- Volunteered at the Special Olympics for science outreach demo titled “Sweet Science” (State College, PA, Summer 2019)
- Diversity in STEM workshop by MRSEC PSU (October 22nd 2018)
- Tutored underprivileged students at Seoul Office of Education in South Korea (Fall 2009)

REFERENCES

Professor Jay T. Groves

Professor of Chemistry
University of California, Berkeley
408A Stanley Hall
Berkeley, CA 94720
jtgroves@lbl.gov, (510) 666-3602

Professor Christine Keating

Shapiro Professor of Chemistry
The Pennsylvania State University
512 Benkovic Building
University Park, PA 16802
cmd8@psu.edu, (814) 863-7832

Professor Philip Bevilacqua

Distinguished Professor of Chemistry and of Biochemistry and Molecular Biology

The Pennsylvania State University

242 Benkovic Building

University Park, PA 16802

pcb5@psu.edu, (814) 863-3812

Introduction

Dr. Saehyun Choi is an Assistant Professor in the Department of Chemistry and Nanoscience at Ewha Womans University. Her research focuses on the physical chemistry of phase transitions in polymers and biomolecules, including coacervates for building artificial cells and biomolecular condensates involved in immune-cell signaling. Her group combines thermodynamic approaches with advanced fluorescence imaging to understand how molecular organization within condensates regulates chemical reactions and cellular signaling.

Before joining Ewha, she was a postdoctoral researcher at the University of California, Berkeley, where she investigated signaling-protein condensates in live T cells using quantitative single-molecule microscopy. She received her Ph.D. in Chemistry from The Pennsylvania State University, where she studied peptide-based coacervates as models of membraneless and prebiotic compartments. Her recognitions include a NASA FINESST Graduate Fellowship, a NASA Astrobiology Early Career Travel Award, and as an elected Chair of the 2024 Gordon Research Seminar on Biointerface Science.